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Comparative analysis of enhanced adsorption and thermal decomposition of oil-borne PFAS using CeO_2 nanoparticles and activated carbon

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ABSTRACT

Per- and poly-fluoroalkyl substances (PFAS) are widely used as surfactants in the oil and gas industry, presenting significant environmental and health risks due to their persistence and mobility. While prevailing research primarily targets PFAS removal from aqueous environments, this study explores the efficacy of nanosized cerium oxide nanoparticles (CeO_2 NPs) and traditional activated carbon (AC) in removing PFAS from organic media via adsorption and thermal degradation. Both adsorbents exhibited robust adsorption capabilities; however, their interaction mechanism with PFAS differ significantly. CeO_2 NPs primarily engage in chemical adsorption with PFAS, whereas AC relies on hydrophobic interactions. Additionally, CeO_2 NPs outperformed AC in thermal degradation experiments, achieving approximately 95 % decomposition of PFOS at 400 °C, compared to only 52 % with AC. Furthermore, the formation of stable Ce−F bonds at high temperatures significantly reduced fluoride ion release from CeO_2 NPs, underscoring their potential to minimize environmental impact. This study is the first to apply both AC and CeO_2 NPs for PFAS removal from organic media and to elucidate their distinct adsorption and thermal decomposition mechanisms, highlighting the superior performance of CeO_2 NPs in environmental remediation of PFAS.

1. Introduction

Per- and poly-fluoroalkyl substances (PFAS) are synthetic chemicals extensively used across various industries, including manufacturing, construction, and firefighting, owing to their exceptional resistance to water, oil, and heat. (Calafat et al., 2019; Anderko and Pennea, 2020) However, due to their persistence and high biological resistance, PFAS can accumulate in both the ecosystem and the human body, increasing the risk of diverse chronic disorders, such as developmental delays and immune system dysfunction. (Ehrlich et al., 2023) Moreover, the robust carbon−fluorine (C − F) bond, with an energy of ~536 kJ/mol, renders PFAS challenging to break down in natural settings and through conventional wastewater treatment processes. (Evich et al., 2022; M.G. Alalm and Boffito, 2022)

PFAS have been integral to the oil and gas industry for over half a century, and they are used extensively in drilling, production aids, and stimulation chemicals. (Murphy and Hewat, 2008; Glüge et al., 2020; Gaines, 2023) This extensive use has led to environmental contamination around oil wells and rigs. (How Colorado is preventing PFAS contamination from the oil and gas industry 2021; Anon., Mapping PFAS "forever chemicals" in oil and gas operations 2021; PFAS in mining and petroleum industry - use, emissions and alternatives 2021; Anon., Oil industry released tons of toxic pollutant 2005) In China, various PFAS compounds, including 6:2 FTAB, PFSA, OBS, and PFCA, have been detected in surface water, sediment, soil, and crude oil products from major oilfields such as Daqing and Dagang. (Xu et al., 2017; Meng et al., 2021; Yao et al., 2022) Intriguingly, PFAS are present in organic solvents, and they tend to exist in more volatile nonionic forms than their ionic forms in water. (Bastow et al., 2022) This behavior suggests that PFAS are likely released into the environment during processes like crude oil refining and transportation, posing significant threats to ecosystems and human health. Despite this, current research has predominantly focused on removing PFAS from environmental matrices such as water, (Gagliano et al., 2020; Yin and Villagrán, 2022; Liu and Sun, 2021; Chen et al., 2024) soil, (Mahinroosta and Senevirathna, 2020; Shahsavari et al., 2021; Bolan et al., 2021) and sewage sludge, (Zhang

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and Liang, 2021; Lakshminarasimman et al., 2021) thereby underscoring a pressing need to develop effective treatment methods for PFAS-contaminated organic media.

Several methods have been proposed to remove PFAS from contaminated aqueous environments, including sequestration technologies such as membrane filtration and adsorption, as well as destruction techniques such as oxidative or reductive destruction and biological treatment. (Lee et al., 2022; Kancharla et al., 2022; Veciana et al., 2022; M.G. Alalm and Boffito, 2022; Zhang et al., 2022) However, most of these approaches face challenges when applied to organic media due to potential issues, such as severe membrane fouling in organic media, the need for water-generated active species, and the requirement for optimal conditions for microbial activity. In contrast, the adsorption method, particularly when combined with thermal treatment, shows promise. This process has been successfully demonstrated using materials like activated carbon (AC), biochar, sludge, and soil heated between 200 and 900 °C. (Duchesne et al., 2020; Wang et al., 2023; Wang et al., 2013; Alinezhad et al., 2022; Xiao et al., 2020) This process involves a sequence of chemical reactions, including radical-mediated initiation, chain propagation, recombination, chain stripping, and termination, ultimately leading to the mineralization of organofluorine compounds into fluoride ions. (Sasi et al., 2021; Aleksandrov et al., 2019) Although AC is favored for PFAS adsorption in aqueous solutions due to its high porosity, specific surface area, and hydrophobic characteristics, (Saeidi et al., 2020; Qu et al., 2009; Park et al., 2020) its effectiveness for removing PFAS in organic media may be limited by competition from oil molecules for adsorption sites. Thus, an adsorbent that specifically interacts with PFAS molecules might be essential for effective adsorption in oil contexts and subsequent thermal degradation.

Cerium oxide (CeO_2), a rare earth metal oxide with a cubic fluorite structure, is known for its versatile applications in fields like catalysis, fuel cells, environmental remediation, medical applications, and energy storage due to its unique chemical and physical properties. (Liu et al., 2024; Zhang et al., 2024; Tu et al., 2024; Cao et al., 2024; Xue et al., 2024) Prior studies have demonstrated that CeO_2 can remove fluoride ions from water by through different mechanisms, including binding at oxygen defects, ion exchange, surface adsorption, and pore filling. (Kang et al., 2017; Pan et al., 2023) These findings confirm the specific interactions and unique affinity between fluoride atoms and CeO_2, suggesting its potential for PFAS removal in organic media. In this study, CeO_2 nanoparticles (NPs) were chosen to capture two legacy PFAS compounds, PFOA and PFOS, in hexadecane, with conventional AC used for comparison. We evaluated the adsorption capacities of both adsorbents towards PFAS, their desorption behaviors, and corresponding thermal degradation efficiencies. Additionally, techniques such as ion chromatography and X-ray photoelectron spectroscopy were employed to detect and analyze free fluoride ions and bonded fluorine species on the adsorbents. This investigation underscores a remediation strategy for PFAS-contaminated organic media, aiming to mitigate the potential release of PFAS and fluoride ions into soil and aqueous environments.

2. Materials and methods

2.1. Chemical reagents

Cerium chloride heptahydrate ($CeCl_3{\cdot}7H_2O$, 99.9 %), sodium hydroxide flakes (NaOH, 97 %), citric acid (≥99.5 %), ammonium acetate (NH_4Ac, 98 %), activated carbon powders (AC, 400 mesh), and hexadecane (98 %) were procured from Makclin Biochemical Co., Ltd. Perfluorooctanoic acid (PFOA, 96 %) and perfluorooctanesulfonic acid potassium salt (PFOS, 95 %) were obtained from Makclin Biochemical Co., Ltd. and Aike Reagent Co., Ltd., respectively. Methanol hypergrade for liquid chromatography LiChrosolv® was purchased from Sigma-Aldrich (St. Louis, MO). Ultrapure water used in solution preparation was supplied from a Millipore Ultrapure Water System. All reagents were used as received without any further purification.

2.2. Adsorption and regeneration experiments

The synthesis of CeO_2 NPs was conducted according to a methodology previously reported, (Xu et al., 2019) with the preparation and characterization procedure detailed in the Supporting Information (Text S1). Batch adsorption experiments were conducted in pre-cleaned polypropylene (PP) vials. Stock solutions of PFOA or PFOS were prepared in hexadecane at concentrations ranging from 0.5 mg/L to 10 mg/L. Adsorbents (AC or CeO_2 NPs) were added at a concentration of 0.5 g/L. The mixed samples were agitated for 24 h to reach adsorption equilibrium, followed by centrifugation at 6000 rpm for 5∼10 min. The residual PFOA or PFOS in the supernatant was extracted with an equal volume of methanol, and their concentrations were determined employing ultra-performance liquid chromatography coupled with a high-resolution orbitrap mass spectrometer (UHPLC-QE-HRMS, Q Exactive, Thermo Fisher Scientific). (Yao et al., 2022) For the desorption tests, spent adsorbents were immersed in 10 mL of methanol, methanol with 0.1 M NaOH (v/v = 1/1), or methanol with 0.1 M NH_4Ac (v/v = 1/1) for a duration of 1 h. After the desorption, the supernatant was then collected to determine the PFAS concentrations. In separate tests, spent adsorbents were immersed in ultrapure water and agitated for 24 h to examine the release behavior of PFAS, which provides insights into the specific interactions between PFAS and the adsorbents.

2.3. Thermal treatment

To completely degrade adsorbed PFAS, the spent samples were placed in a 75 mL sealed borosilicate glass container and subjected to thermal decomposition in a preheated muffle furnace, following the procedures outlined in Xiao's prior research. (Xiao et al., 2020; Sasi et al., 2021) The temperature for thermolysis ranged from 100 to 400 °C, with durations spanning 5 to 40 min. Following thermal treatment, samples were immediately cooled in a fume hood. Either ultrapure water or methanol with 0.1 M NaOH was then added to the resulting product, and the mixture was ultrasonicated at room temperature for 30 min. The residual concentrations of PFOA and PFOS in methanol were measured by UHPLC-QE-HRMS, and fluoride ions released into the water were detected by ion chromatography (IC, Dionex Aquion RFIC, Thermo Fisher Scientific).

3. Results and discussion

3.1. Materials characterization

The TEM image revealed nanosized CeO_2 particles synthesized via the hydrothermal method (Fig. 1a). The inserted HRTEM image indicated an interplanar distance of 0.313 nm, corresponding to the (111) planes of CeO_2. (Kang et al., 2017; Xu et al., 2019) As shown in Fig. 1b, XRD analysis confirmed the presence of characteristic peaks at 28.6°, 33.1°, 47.5°, 56.4°, 59.2°, 69.5°, 76.8°, and 79.1°, each corresponding to the (111), (200), (220), (311), (222), (400), (331), and (420) crystal planes, respectively (JCPDS 43–1002). These sharp peaks are indicative of well-formed crystalline particles with a cubic fluorite-type structure, aligning with findings reported in prior literature. (Parimi et al., 2019; Michalska et al., 2021) In addition, the synthesized CeO_2 NPs displayed a specific surface area (SAA) of ∼ 70 m^2/g, accompanied by a broad pore size distribution ranging from 15 to 50 nm. This could be attributed to the aggregation of small-sized nanoparticles, as observed in Figs. 1c and d. In contrast, commercial AC particles exhibited an aggregated morphology and a significantly larger SAA exceeding 900 m^2/g, characterized by a rich micro/mesoporous structure (Figure S1 and S2).

3.2. Adsorption performance and removal mechanisms

Both CeO_2 NPs and AC demonstrated high removal efficiencies (>90 %) in organic media for PFOA across initial concentrations ranging from

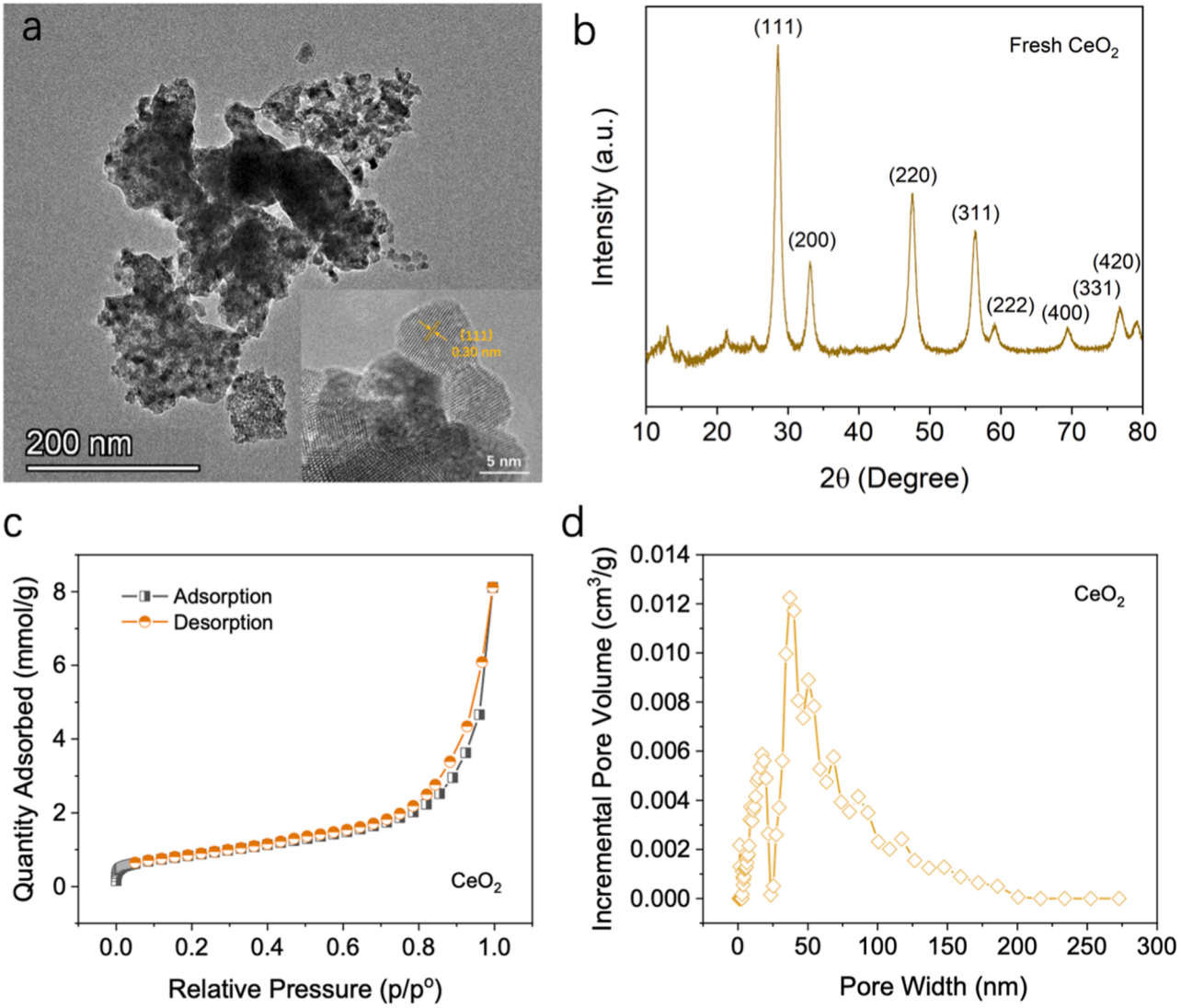


Fig. 1. Characterization of the prepared CeO_2 NPs: (a) TEM image of CeO_2 NPs with an inserted image indicating crystal information, (b) XRD pattern, (c) BET adsorption-desorption isotherm curves, and (d) pore size distribution.

0.5 to 10 mg/L, with an adsorbent dosage of 0.5 g/L, as shown in Fig. 2a. A similar trend was observed for the removal of PFOS (Fig. 2b). To rule out the possibility that the oil medium may preferentially adsorb the pollutants rather than the adsorbents themselves, we conducted PFOA and PFOS removal experiments in a water medium using both adsorbents. The results demonstrated that both AC and CeO_2 NPs effectively removed over 96.5 % of PFOA and PFOS across a range of initial concentrations, from 0.5 ppm to 10 ppm (Figure S3). To further compare the adsorption performance of AC and CeO_2 NPs, varying dosages of both adsorbents were tested in hexadecane with PFOA. As illustrated in Figure S4, CeO_2 NPs demonstrated superior removal performance compared to AC at lower dosages of 0.1 and 0.3 g/L. This advantage became more pronounced at higher PFOA concentrations. For instance, CeO_2 NPs removed over 98 % of 1 ppm PFOA, whereas AC achieved approximately 80 % removal. In addition, the adsorption kinetics curves for PFOA and PFOS in oil media indicated that the rapid adsorption predominantly occurred within the first two hours (Figure S5), with no significant difference in concentration changes induced by AC and CeO_2 NPs. While the large SSA, appropriate pore size, and surface chemistry of adsorbents are crucial for PFAS removal in aqueous environments, (Pauletto and Bandosz, 2022) our findings suggest that the importance of SSA and pore size may be reduced in organic media considering the apparent difference between CeO_2 and AC in terms of SSA and pore size.

To investigate the removal mechanism of PFAS by these adsorbents, the used AC and CeO_2 NPs were immersed in ultrapure water, and the concentration variations of PFOA and PFOS were monitored over time as secondary releases from the adsorbents (Fig. 2c and d). Notably, there was a significant release of PFOA (~70 %) and PFOS (~100 %) from AC into the water after 24 h. This behavior may be attributed to the removal process being predominantly driven by hydrophobic interactions in organic media, which concentrate PFAS molecules within the porous channels and on the outer surfaces of the AC. The weak chemical affinity between PFAS and AC facilitates their release into the water.

In contrast, CeO_2 NPs exhibited a modest initial release of PFAS, followed by maintaining only approximately 15 % release efficiency for PFOA and 20 % for PFOS over the following 24 h. FTIR spectra and elemental analysis were used to assess the residual PFOA and PFOS on CeO_2 NPs. The FTIR spectra of pristine CeO_2 NPs displayed two characteristic bands at 3387 and 686 cm^{-1}, corresponding to the O—H vibrations in absorbed water and the stretching frequency of Ce-O, respectively (Figure S6). (Wang et al., 2010) After PFOA and PFOS adsorption, a distinct peak appeared in the range of 1100–1240 cm^{-1}, which is attributed to the vibrations of the -CF_3 and -CF_2- groups in organic fluorine. (Zhang et al., 2013; Chen et al., 2017) Additionally,

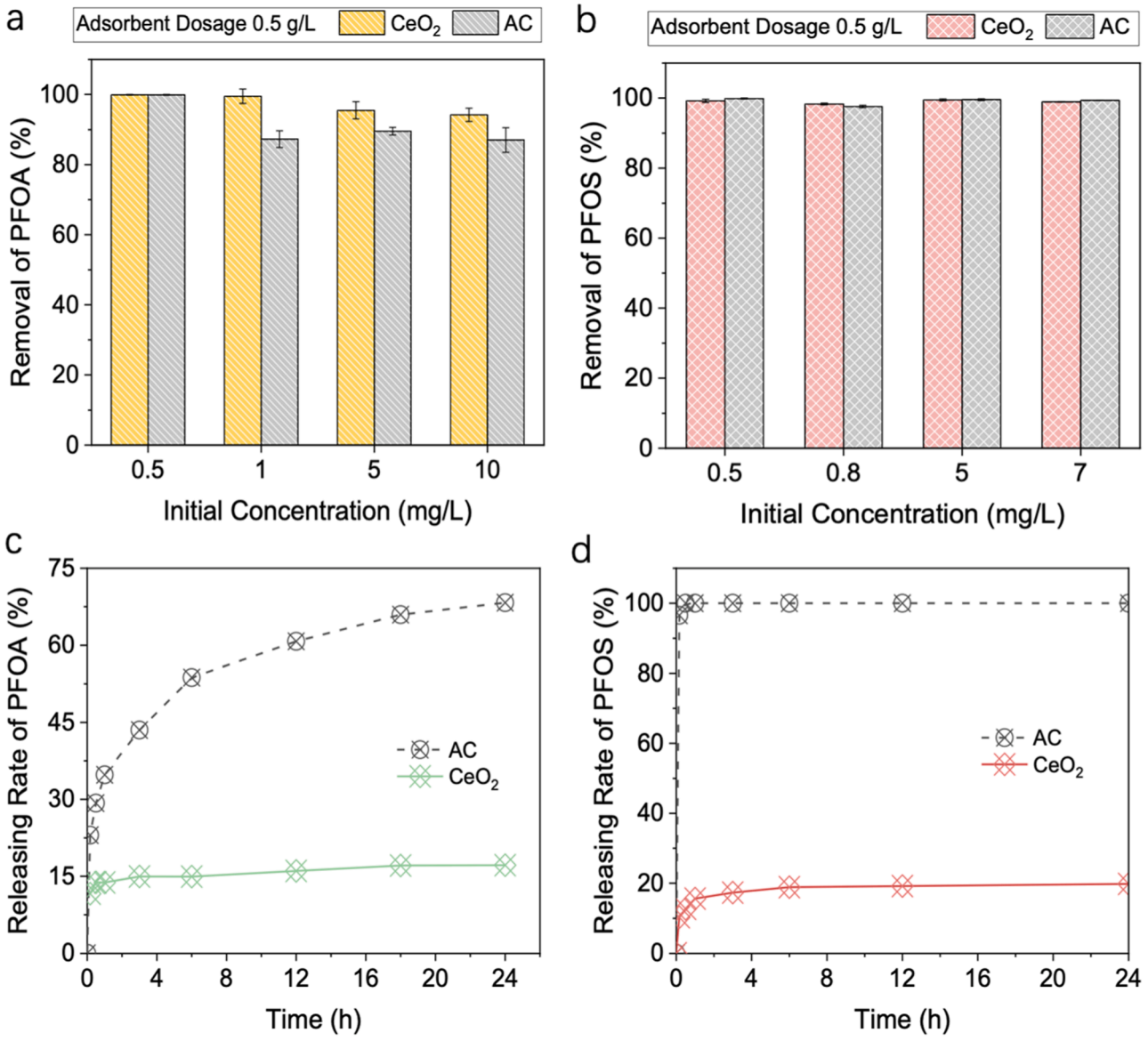


Fig. 2. Comparison of the removal performance of CeO_2 NPs and AC powder for (a) PFOA and (b) PFOS in n-hexadecane; releasing rate of (c) PFOA and (d) PFOS on spent CeO_2 NPs and AC powder in ultrapure water.

elemental mapping images demonstrated the presence of fluorine onto CeO_2 NPs after PFAS adsorption (Figure S7 and S8). High-resolution TEM image further indicated that the lattice distances were 0.307 nm, consistent with the (102) plane of the hexagonal CeF_3 (Figure S9). (Camus-Génot et al., 2024) We hypothesize that the strong affinity between CeO_2 NPs and PFAS results in the relatively stable anchoring of the pollutants on the adsorbent with the minimal release of PFAS. In summary, the superior adsorption stability provided by CeO_2 NPs not only reduces the volatility of PFAS in organic media but also decreases the secondary migration of PFAS when CeO_2 NPs are used in aqueous environments.

We further explored the adsorption and desorption mechanisms by immersing PFOA-laden CeO_2 NPs and AC in MeOH, a solvent recognized for its ability to disrupt hydrophobic interactions. (Siriwardena et al., 2021) As shown in Fig. 3a, MeOH successfully extracted 13–50 % of PFOA from CeO_2 NPs, with extraction efficiency varying based on the initial concentration. A higher extraction efficiency suggests the presence of a thicker and less stable hydrophobic adsorption layer around the adsorbents, likely influenced by the limited SSA of CeO_2 NPs. Additionally, the inclusion of 0.1 M NH_4AC or NaOH in MeOH significantly enhanced the extraction efficiency of PFOA to 90 % and 100 %, respectively. In contrast, PFAS was more readily desorbed from the porous channels and outer surface of AC, predominantly due to weak physical adsorption (i.e., hydrophobic interactions) in organic media. This observation underscores the distinct adsorption and desorption behaviors driven by the structural and chemical characteristics of the two adsorbents.

To further elucidate the mechanism of PFAS removal by CeO_2 NPs, XPS analysis was conducted. In high-resolution Ce3d XPS spectra of the fresh CeO_2 NPs, two oxidation states (i.e., Ce^{3+} and Ce^{4+}) were identified, with a Ce^{3+}/Ce^{4+} ratio of 0.18 (Fig. 3c). The O1s spectra revealed four distinct chemical states at 529.2 eV (Ce^{4+}-O^{2-}), 529.7 eV (Ce^{3+}-O^{2-}), 531.4 eV (-OH), and 533.3 eV (adsorbed H_2O). The ratio of Ce^{4+}-O^{2-}/Ce^{3+}-O^{2-}, which represents the content of lattice oxygen, was determined to be 1.28. Following PFOA adsorption, the Ce^{3+}/Ce^{4+} ratio increased to 0.21, indicating the partial conversion of $Ce^{4+}$to Ce^{3+}. (Wang et al., 2019; Chen et al., 2021) Concurrently, the lattice oxygen content decreased to 1.06, suggesting that the fluorine atoms in PFOA may replace the lattice oxygen in CeO_2, leading to the release of two electrons and the subsequent reduction of $Ce^{4+}$to Ce^{3+}. (Schmitt et al.,

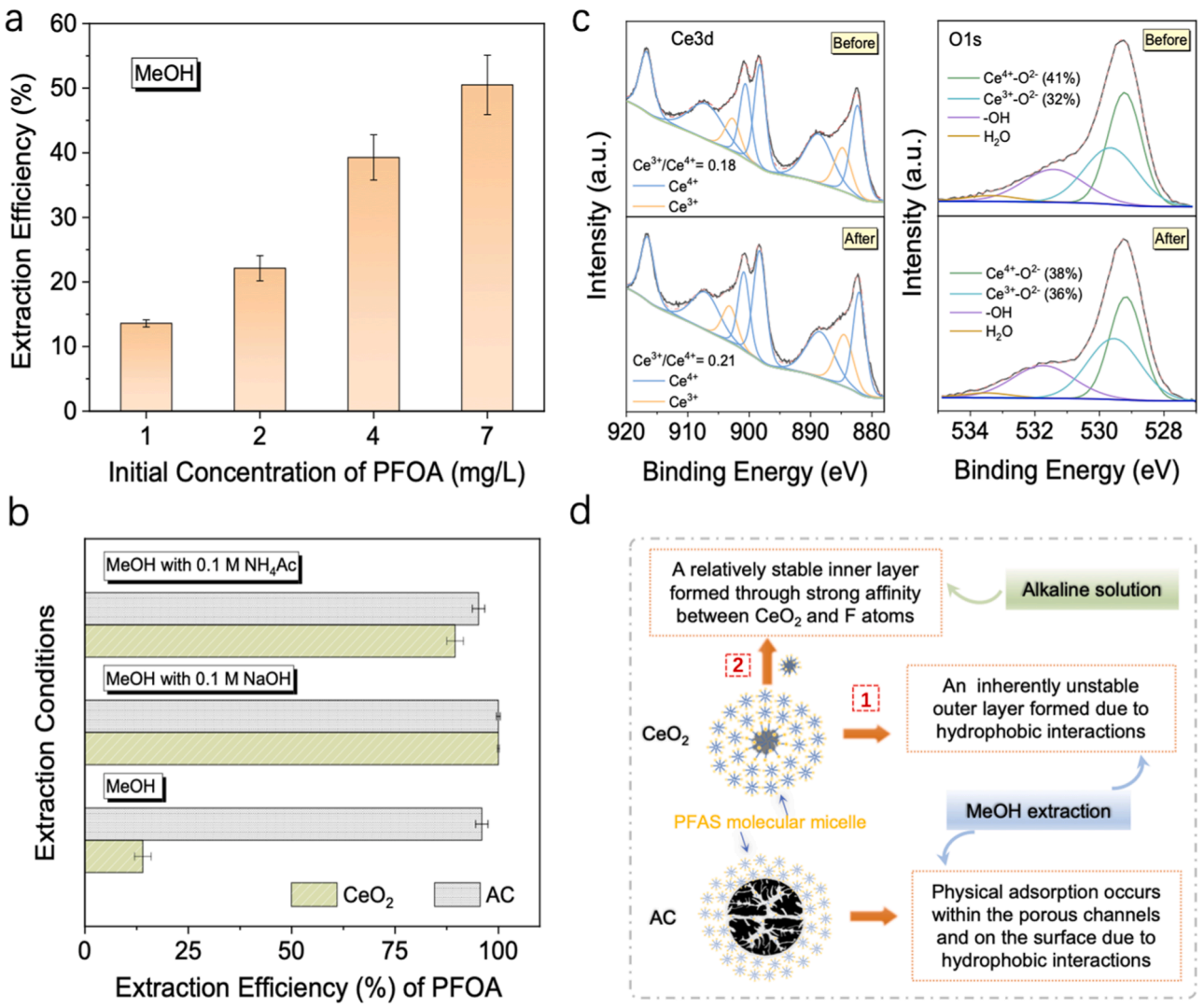


Fig. 3. Desorption behaviors and proposed adsorption mechanism of CeO_2 NPs: (a)-(b) recovery of PFOA from CeO_2 NPs and AC powder by different extraction conditions; (c) high-resolution XPS spectra for Ce 3d and O 1 s of CeO_2 NPs before and after PFOA adsorption; (d) schematic illustration of the removal mechanisms of PFOA molecules by CeO_2 NPs and AC.

2020; Sayle et al., 2016; Campbell and Peden, 2005)

Based on the above findings, we propose a two-step adsorption model for PFAS onto CeO_2 NPs in organic media (Fig. 3d). In the first step, a portion of PFAS molecules forms strong chemical bonds with the surface of CeO_2 NPs. This interaction primarily involves the exchange or sharing of electrons between PFAS and the CeO_2 surface, resulting in stable adsorption that can only be disrupted in an alkaline solution. After the desorption process, the Ce^{3+}/Ce^{4+}ratio decreased from 0.21 during adsorption to 0.17, approaching the value observed in the raw CeO_2 (Figure S10). Meanwhile, the Ce^{4+}-O^{2-}/Ce^{3+}-O^{2-} ratio significantly increased to 4.3. This result suggests the partial conversion of Ce^{3+} to Ce^{4+} and the formation of lattice oxygen, which is beneficial for performing multiple adsorption-desorption cycles effectively. As shown in Figure S11, PFOA removal efficiency remained above 96 % during the first three cycles, followed by a slight decrease to around 90 %, likely due to the loss of adsorbent during the recycling process. When the available adsorption sites on the CeO_2 surface are saturated with PFAS molecules, the excess PFAS in the organic media may form molecular micelles. These micelles then assemble an outer, less stable adsorption layer around the inner chemically bonded layer through hydrophobic interactions. These weaker physical interactions facilitate the easier extraction of PFAS using MeOH. Thus, the adsorption of PFAS molecules on CeO_2 NPs in organic media is characterized by a combination of both chemical and physical interactions.

3.3. *Thermal decomposition efficiency and mechanisms*

To mitigate the environmental impact of PFAS-contaminated adsorbents, thermal treatment at moderate temperatures has proven to be an effective strategy for managing such contaminated solid wastes. (Vidonish et al., 2016) It has been reported that AC can significantly influence the thermolysis behavior of adsorbed PFAS compared to PFAS powders alone, enhancing both decomposition and defluorination efficiencies at temperatures below 400 °C. (Sasi et al., 2021) To assess the thermolysis performance, both PFOA-laden and PFOS-laden AC, as well as CeO_2 NPs, were placed in a closed container and subjected to heating in a muffle furnace at predetermined temperatures (100, 150, 200, 300, 350, or 400 °C) for varying durations from 5 to 30 min. As shown in Fig. 4, the efficiency of thermal decomposition was markedly affected by the heating temperature setting. Notably, temperatures below 100 °C were ineffective in degrading PFOA, as its decomposition threshold temperature was measured to be about 150 °C (Fig. 4a). (Wang et al., 2015) An increase in the heating temperature from 150 to 300 °C led to a substantial increase in the thermal decomposition of PFOA adsorbed on

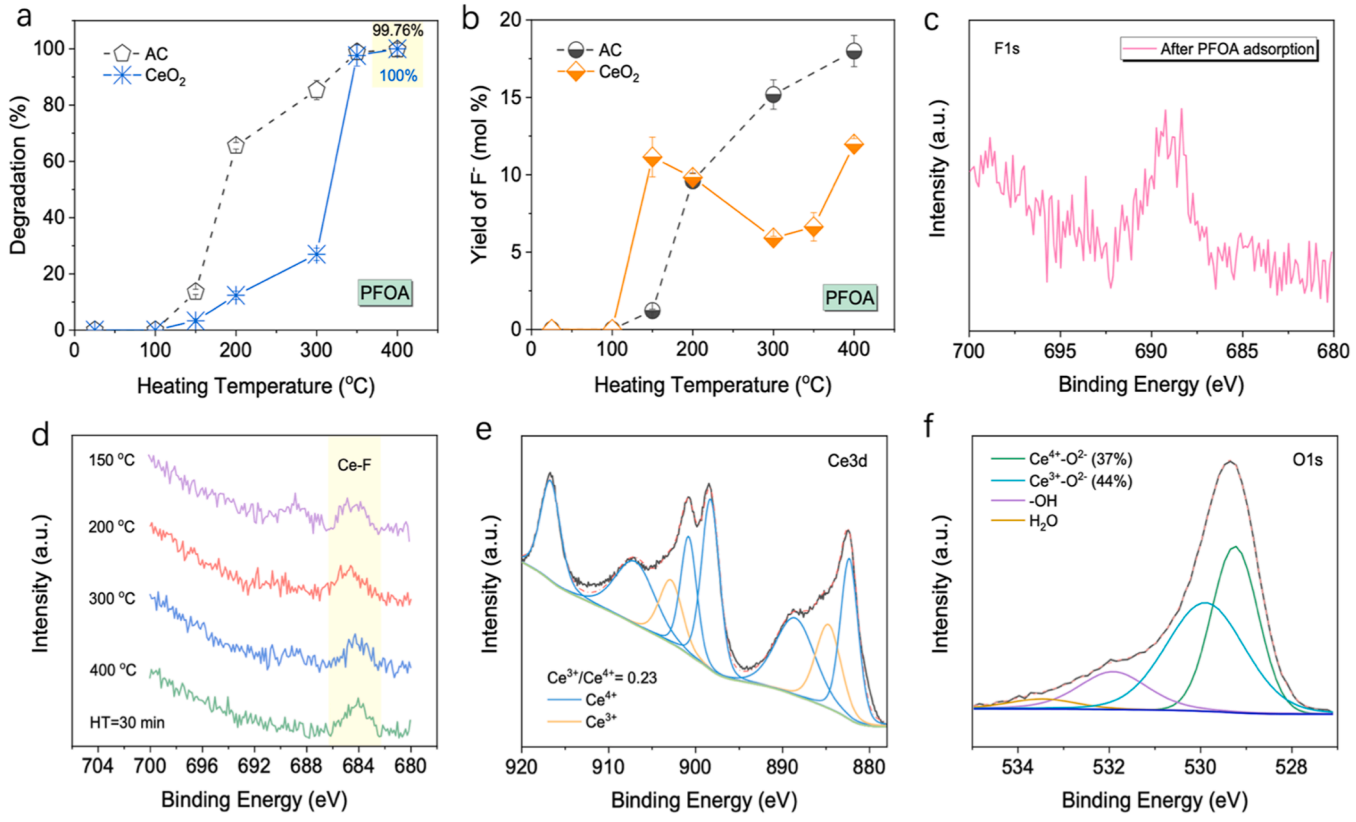


Fig. 4. (a) Decomposition of PFOA adsorbed on two adsorbents (0.1 g AC laden with 3.3×10^{-7} mol PFOA; 0.1 g CeO_2 NPs laden with 3.8×10^{-7} mol PFOA), and (b) measured yield of F^- after 30 min-thermal treatments at different temperatures; (c) High-resolution XPS spectra of F 1 s from PFOA adsorbed on CeO_2 NPs; (d-f) High-resolution XPS spectra of F1s, Ce 3d, and O 1 s region of CeO_2 NPs after undergoing thermal treatment at different temperatures.

AC, with the efficiencies increasing from 13 % to 85 %. At temperatures exceeding 350 °C, >99 % of PFOA was decomposed within 30 min. The thermal decomposition process involves various stages, including melting, evaporation, and multistep chemical reactions, primarily yielding organofluorine species, as recently explored by Xiao's group. (Xiao et al., 2021; Alinezhad et al., 2023) Concurrently, there was a notable increase in fluoride ion production, which could be subsequently collected using ultrapure water under sonication. The yield of fluoride ions varied from 1.2 mol% at 150 °C to 18 mol% at 400 °C, as shown in Fig. 4b. This increase in both degradation and defluorination efficiencies suggests the cleavage of both C–C and C–F bonds during the thermal treatment, which is analogous to the thermal degradation process of PFAS adsorbed in water. (Xiao et al., 2020)

The thermolysis behavior and pathways of PFOA adsorbed on CeO_2 NPs exhibit notable differences compared to its adsorption on AC. Specifically, the volatility of PFOA on CeO_2 is reduced, largely because it predominantly adsorbs through chemical interactions rather than hydrophobic interactions. Consequently, the thermolysis process at 150 °C is primarily characterized by a defluorination mechanism, leading to the generation of >10 mol% of washable free fluoride ions from CeO_2 NPs (Fig. 4b). If PFOA adsorbed on CeO_2 NPs follows a similar thermolysis behavior and pathway as that on AC, both degradation and defluorination would typically occur concurrently. In other words, when exposed to thermal energy, both C—C and C-F bonds are equally activated and cleaved. In fact, the chemical interaction between CeO_2 and F atoms, originating from PFOA's hydrophobic tails, may reduce the

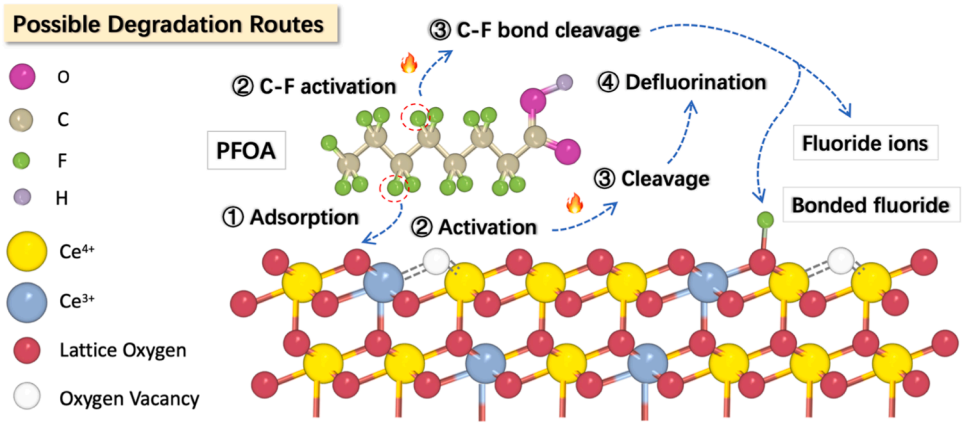


Fig. 5. Possible PFOA degradation routes on the CeO_2 NPs.

energy of adjacent C − F bonds at the CeO_2-F interaction sites (routes 1 and 2 in Fig. 5)). This interaction results in a preferential cleavage of C − F bonds over C − C bonds at lower temperatures (route 3), showcasing a unique thermolysis pathway facilitated by CeO_2.

As the temperature is raised to 200 and 300 °C, there is a noticeable reduction in the yield of fluoride ions (Fig. 4b), despite an increase in PFOA degradation efficiency (Fig. 4a). This reduction could be attributed to the enhanced binding affinity of the generated fluoride ions with CeO_2 NPs at higher temperatures. (Sun et al., 2021) When the temperature is further elevated to 400 °C, complete degradation of PFOA is observed, alongside a concurrent increase in the yield of fluoride ions due to the higher thermal energy input. This observation is particularly noteworthy considering the thermal conductivity properties of the materials involved. Porous AC has been demonstrated to facilitate the thermal decomposition of PFAS, which has been attributed to its higher thermal conductivity (0.4–1.36 W/(m·K)) compared to air (<0.032 W/(m·K)) at temperatures ≤100 °C. (Sasi et al., 2021) In contrast, CeO_2 exhibits even higher thermal conductivity, measuring between 7 and 8 W/(m·K). (Suzuki et al., 2019) Therefore, the superior thermal conductivity of CeO_2, coupled with specific chemical interactions between PFAS and CeO_2, significantly enhances the efficiency of thermal degradation.

To elucidate the interaction between CeO_2 and fluorine species during thermal treatment, the CeO_2 NPs subjected to thermolysis were repeatedly washed with ultrapure water to eliminate any loosely adsorbed fluoride ions. The dried samples were then characterized for elemental compositions and chemical states using TEM and XPS. Figure S12 displayed the elemental mapping of the CeO_2 NPs after thermolysis, revealing the presence and distribution of fluorine throughout the nanoparticles. High-resolution XPS spectra indicated a significant shift in the main peak of F 1 s from PFOA-laden CeO_2 spectra, moving from 688.5 eV to 684.2 eV after thermal treatment (Figs. 4c and d). This shift suggests the formation of Ce−F bonds, (Han et al., 2020) aligning with the peak position of commercially available CeF_3 (Figure S13). Further XPS analysis of the Ce 3d and O 1 s spectra revealed an increased Ce^{3+}/Ce^{4+} ratio (0.23) and a decreased Ce^{4+}-O^{2-}/Ce^{3+}-O^{2-} ratio (0.84) (Figs. 4e and f). These changes suggest that the generated fluorine may replace lattice oxygen and yield more Ce^{3+}, thereby supporting the successful binding of fluorine and CeO_2. To further elucidate the reduction mechanism of the Ce element, Raman spectroscopy was also conducted. As reported, the ratio of the peak intensities at 598 and 462 cm^{-1} is positively correlated with the concentration of oxygen vacancies in CeO_2. (Pan et al., 2023; Wang et al., 2017) Figure S14 showed that this ratio increased from 1.40 % to 1.82 % after thermal treatment, indicating the formation of additional oxygen vacancies. It is well established that the amount of Ce^{3+} is directly related to the concentration of oxygen vacancies in CeO_2, with the creation of an oxygen vacancy corresponding to the generation of two Ce^{3+}ions. (Bêche et al., 2008) This finding implies that more Ce^{3+} was generated after the thermolysis of the PFOA-laden CeO_2. Additionally, XPS analysis provided insights into the transition from organofluorine to inorganic fluoride across the temperature range of 150–400 °C. The high-resolution XPS spectra of F 1 s at 150 °C showed two distinct peaks (Fig. 4d). The peak representing organofluorine (C − F) at approximately 689 eV gradually diminished in intensity with increasing temperature, while the peak associated with inorganic fluoride grew more pronounced. By 400 °C, the organofluorine peak had vanished entirely, marking the complete degradation of PFOA, as corroborated by the observations in Fig. 4a.

Unlike the thermolysis behavior of PFOA, PFOS exhibits less volatility and has a higher decomposition temperature of 425 °C in O_2-rich environments and 450 °C in N_2 atmosphere. (Wang et al., 2015; Xiao et al., 2023) Consequently, PFOS adsorbed on AC, biochar and soil can undergo almost complete degradation when exposed to temperatures exceeding 500 °C. (Wang et al., 2023; Alinezhad et al., 2022; Xiao et al., 2020) However, lowering the temperature to 400 °C results in a significant reduction in degradation efficiency, dropping to only 50–60 % after a thermolysis duration of 15–30 min (Fig. 6a). These findings highlight the challenge of achieving high degradation efficiency for PFOS at moderate temperatures. Similar to the thermolysis behavior observed for PFOA-laden AC, PFOS adsorbed on AC also demonstrated an increase in both degradation and the yield of F^- as the temperature was raised from 100 to 400 °C (Fig. 6b). However, due to its lower volatility, PFOS exhibited a lower extent of defluorination compared to PFOA.

The thermolysis behavior of PFOS on CeO_2 NPs displays similarities with that of PFOA on CeO_2 NPs. Specifically, PFOS on CeO_2 NPs showed a consistently lower degradation efficiency within the temperature range of 150 to 300 °C. However, as the temperature was raised to 400 °C, approximately 94.4 % of the PFOS underwent thermal degradation, accompanied by a slight increase in the yield of F^-. High-resolution XPS spectra of the heat-treated CeO_2 NPs unveiled the existence of non-removable inorganic fluorine species at a binding energy of around 684 eV, as shown in Fig. 6c. At a temperature of 400 °C, the enhanced peak corresponding to F 1 s suggests a more pronounced fluorine bonding configuration. Moreover, TEM and elemental mapping images further corroborated the presence of chemically bonded fluorine on the surface of the CeO_2 NPs following the thermal treatment (Figure S15). These findings contribute to a deeper understanding of the thermolysis behavior of PFAS on CeO_2 NPs, highlighting the distinct pathways and efficiencies of thermal degradation and defluorination across different adsorbents and PFAS compounds.

The duration of the thermolysis process was also examined to assess its impact on thermal decomposition. Both PFOA on AC and CeO_2 NPs exhibited rapid decomposition, achieving nearly 100 % degradation efficiency within 10 min (Fig. 7a-b). However, the yield of fluoride ions demonstrated a notable increase after 20 min of heating, with no significant further increase observed as the heating time was extended to 40 min (Fig. 7c). Conversely, the thermal decomposition of PFOS on AC reached approximately 50 % within 20 min (Fig. 7a). Extending the duration beyond this point did not result in a significant increase in decomposition. In this scenario, the defluorination process only showed a modest increase of 5 mol% (Fig. 7d).

In contrast, the introduction of CeO_2 NPs has a significant impact on the kinetics of thermal decomposition of PFOS, leading to remarkable improvements in degradation efficiency. The degradation efficiencies of PFOS reached 58 % within 10 min and 84 % within 20 min (Fig. 7b). Remarkably, when the heating duration was prolonged to 30 and 40 min, the degradation efficiency escalated to approximately 95 %, a level considerably higher than those previously reported in the literature. (Wang et al., 2023; Alinezhad et al., 2022; Xiao et al., 2020) Moreover, high-resolution F 1 s spectra demonstrated that the intensity of the peak associated with bonded fluorine progressively increased with prolonged heating duration (Figure S16). This observation suggests a continual formation of bonded fluorine species over time, underscoring the dynamic nature of the thermolysis process and its dependence on both the duration and temperature of heating.

Additionally, cycling experiments for PFOA adsorption and degradation were conducted to assess the stability and reusability of CeO_2 NPs after thermal treatment. As demonstrated in Figure S17, the adsorbent retained approximately 80 % PFOA removal efficiency even after five cycles, with the degradation efficiency consistently reaching 100 %. The reduction in removal capacity is attributed to the partial occupation of adsorption sites on the CeO_2 NPs by F atoms during the heating process (Figure S18). Furthermore, the XRD pattern revealed a significant decrease in the characteristic peak intensities of CeO_2 NPs after five cycles, indicating diminished crystallinity due to bonded F atoms (Figure S19). High-resolution XPS spectra in Figure S20 further revealed a continued increase in the Ce^{3+}/Ce^{4+} ratio (0.25) and a significant decrease in the Ce^{4+}-O^{2-}/Ce^{3+}-O^{2-} ratio (0.17) compared to the values in the initially heated CeO_2 (Figs. 4d and e). These findings indicate that during the cycling experiments, fluorine atoms further replaced lattice

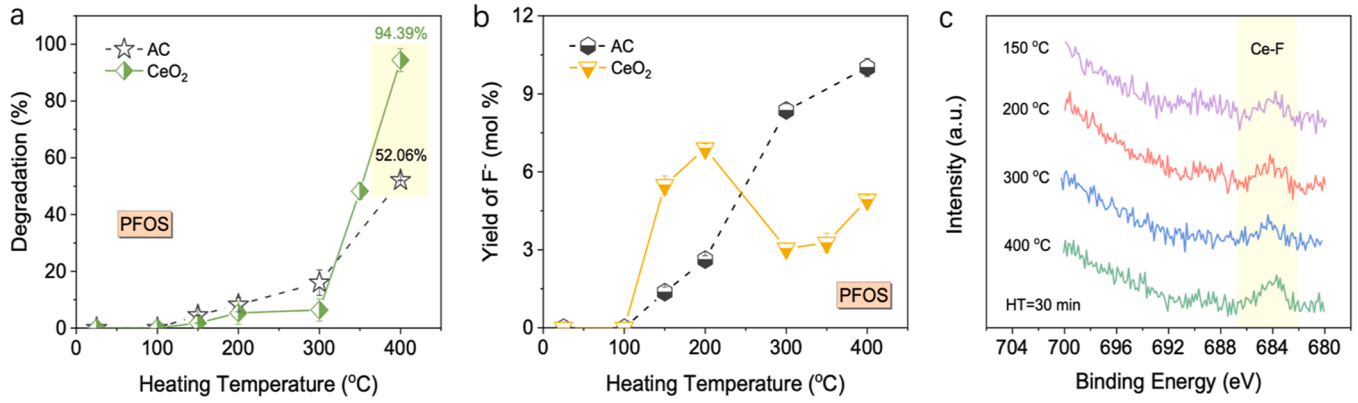


Fig. 6. (a) Thermal decomposition of PFOS adsorbed on two adsorbents (0.1 g AC laden with 3.1×10^{-7} mol PFOS; 0.1 g CeO_2 NPs laden with 3.7×10^{-7} mol PFOS), and (b) measured yield of F^- after 30 min-thermal treatment at different temperatures; (c) High-resolution XPS spectra of F1s region of PFOS-laden CeO_2 NPs after undergoing thermal treatment at different temperatures.

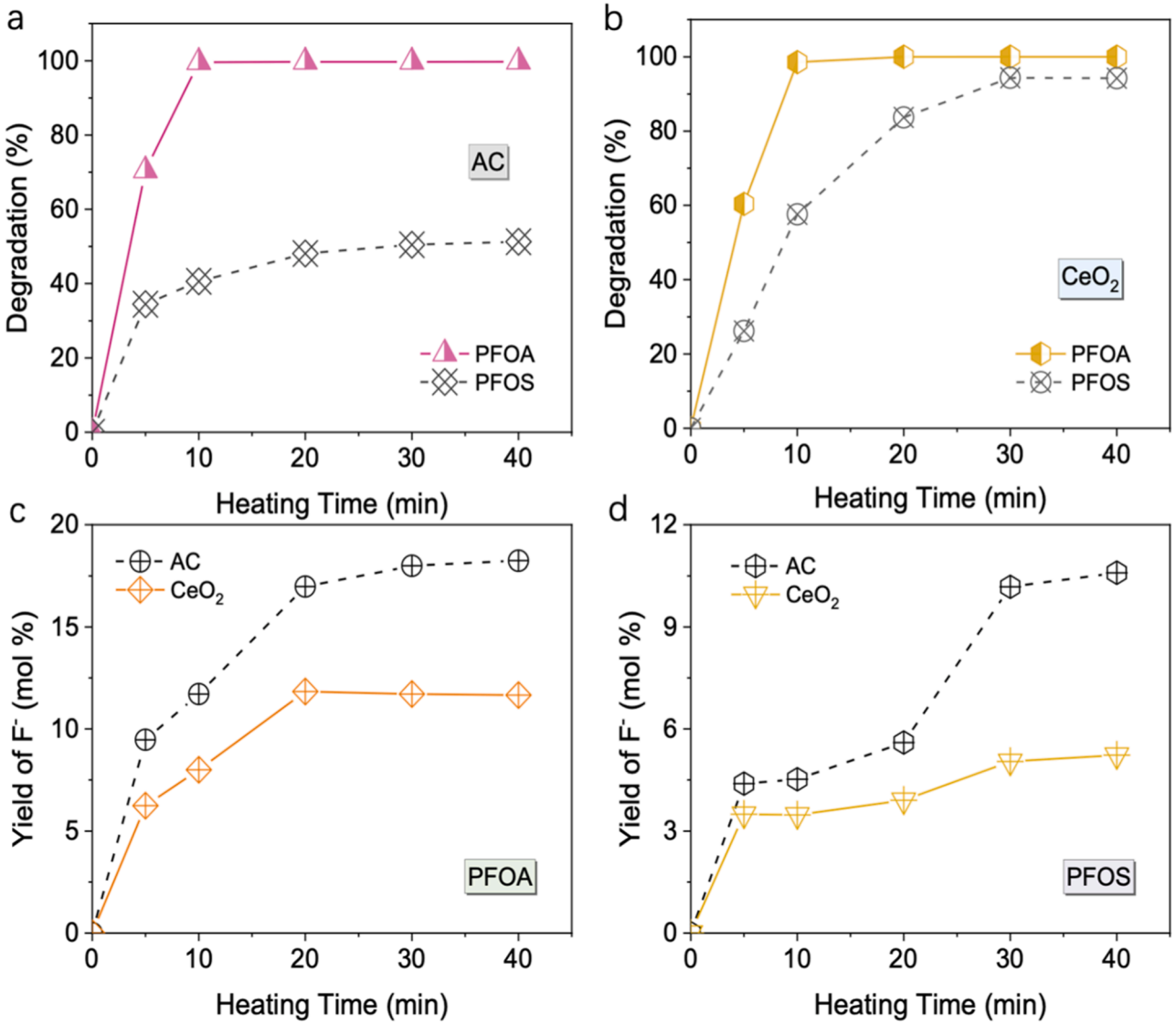


Fig. 7. Thermal decomposition of PFOA and PFOS adsorbed on (a) CeO_2 NPs and (b) AC powder, and (c)-(d) measured yield of F^- as a function of heating time at 400 °C.

oxygen, resulting in the generation of additional Ce^{3+} ions.

4. Conclusions

CeO_2 NPs have demonstrated significant efficacy in removing PFOA and PFOS from organic media, leveraging a synergistic combination of chemical and hydrophobic interactions. Notably, during the thermolysis process, the chemical bonds formed between CeO_2 and PFAS are crucial in limiting the volatilization of these compounds, effectively restraining their transition to the gas phase while facilitating the activation and cleavage of C − F bonds. Furthermore, the high thermal conductivity of CeO_2 ensures efficient energy utilization, enhancing the breakdown of molecular chains during thermal degradation. Although quantifying the bonded fluorine on CeO_2 NPs post-decomposition presents challenges, XPS spectral analysis has confirmed the presence of chemically bonded fluoride after the decomposition of both PFOS and PFOA. Distinct from traditional porous carbon-based adsorbents, CeO_2 NPs offer several notable advantages. They enable highly efficient thermal decomposition of PFAS, particularly PFOS, at temperatures below 400 °C. Furthermore, the immobilization of fluorine species on the surface of CeO_2 NPs and the consequent reduction in the release of free fluoride ions significantly diminish the potential ecological impact. This capability positions CeO_2 NPs as a superior alternative for the advanced treatment of PFAS-contaminated environments, combining high decomposition efficiencies with environmental safety.

CRediT authorship contribution statement

Linlin Zang: Writing – original draft, Methodology, Investigation, Data curation. **Chaoqun Zhu:** Investigation, Data curation. **Meng Zhang:** Software, Investigation. **Yufei Shu:** Software, Methodology. **Xun Liu:** Methodology, Investigation. **Zhongying Wang:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

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